

Validation of CFD models of urban microclimates under high temperature and humidity conditions during daytime heatwaves in dense low-rise areas

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ABSTRACT

In the midst of the ongoing climate crisis, there is a growing need to study the effects of climate on humans, especially those of the microclimates that surround humans and their activities. Various models have been developed to simulate microclimates, but simulation models for densely populated areas experiencing heat waves have shown an issue with simulating temperatures that were too high compared to actual measurements. To overcome these limitations, this study aimed to develop and validate CFD model based on STAR-CCM+ in order to more accurately simulate the microclimate of dense residential areas. To improve the realistic simulation capability found in existing models, the model was designed to include solar radiation, humidity, ground radiation, longwave radiation exchange, heat storage and evaporation due to various urban surfaces, evapotranspiration from vegetation, convection, and heat transfer from buildings. For validation, comparisons were made between the simulation results and the actual measurements during this time. The results showed that the R^2 value of the model developed in this study increased from 0.86 to 0.91, indicating that the model was able to reproduce the measured results of the target site well. To avoid overfitting the model, the measurement results of the Han River AWS, another AWS operated by the Korea Meteorological Administration near the target site, were input. It was confirmed that the model reproduced the actual measurements of five locations in the target site well. The R^2 of temperature and humidity ranged from 0.96 to 0.98, and wind speed ranged from 0.64 to 0.81.

1. Introduction

Urbanization is also causing adverse impacts on the global environment [1]. In particular, changes in urban climate (the local climate within cities) and climate change caused by the deterioration of the overall global environment, have been a concern since the early 2000s [2]. In the field of urban climatology, thanks to advances in computing, there has been an increase in recent years in the number of research methods that utilize computer simulations. Computer simulations can be applied to easily evaluate response strategies under various scenarios. They have the advantage of being able to analyze any location within the computational domain of a computer, not just specific measurement points based on existing weather stations. Computational fluid dynamics (CFD), a prominent example of these simulation techniques, is gaining attention as a meaningful method for analyzing the urban heat island phenomenon since it can perform a range of analyses and set up a virtual space that implements specific conditions [3].

The main advantage of simulation studies is that they allow

comparison with the aforementioned observational studies [4]. Observational studies performed through field observations or automatic weather stations can obtain accurate data, but they have the disadvantage of being limited to a specific location. On the other hand, simulation studies can directly analyze points if they are within the computational domain, and even make future predictions. However, they present poor explanatory power and require additional validation of the CFD model. Comparative studies that combine these two methods can raise the validity of the model and allow for analysis and prediction at various points. In fact, this combination of simulation and observational studies has been widely used to measure LST in the context of the urban heat island phenomenon, as well as to predict building heat loads from an urban microclimate perspective (see Table 1).

CFD analysis techniques are increasingly being used in architecture, urban planning, and design to address air and thermal environmental issues [68–70]. To simulate heat waves, insolation is additionally modeled, which can result in excessively high surface temperatures and atmospheric temperatures that exceed actual measurements. This is

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because there is energy input (shortwave radiation) from insolation, but the process of losing this energy is not described in detail in simulations, resulting in an energy balance at an excessively high temperature point. The results of most CFD simulations of heat waves exceed the actual measured temperatures by more than 6–8 °C, with summer peak temperatures exceeding 40 °C [41,68]. It can be inferred from previous studies that it is not easy to adequately represent changes in the thermal environment of an external space using CFD, but there are methods to solve this problem such as by not modeling insolation [19,70] or by attempting to perform a steady analysis based on a specific point of time [69].

This study noted that climate change is causing climatic shifts in temperate regions. According to the Korea Meteorological Administration (2022) [71], summers in South Korea are growing longer along with climate change, the average summer temperature has increased by 1.6 °C compared to 30 years ago, and the number of days of precipitation has increased to 22.1, which is 28 % higher than the 30-year average of 17.3 days. This shows that South Korea, which is located in a temperate climate region, is transitioning to a subtropical climate with climate change. This phenomenon is not unique to South Korea, but is a problem faced by many countries and/or cities in formerly temperate climates. Unfortunately, according to Arnfield (2003) [72], compared to temperate climates, subtropical regions have been less studied in terms of urban microclimates, including urban heat islands. In particular, there are fewer studies on regions that are transitioning from temperate to subtropical climates since the phenomenon is just beginning.

In this respect, the present study is distinct in that it focuses on the development of a microclimate prediction model capable of reproducing elevated temperature and humidity environments in temperate regions that are undergoing changes. The challenge of using CFD-based modeling to improve the reproducibility and explanatory power of low-to medium-rise, ultra-high-density areas with complex urban/architectural morphologies is considerable, as is the challenge of using CFD-based modeling to improve the reproducibility and explanatory power of hot and humid climatic environments with a complex combination of climatic and environmental factors. In fact, as pointed out by Keller Sullivan Oliveira Rocha et al. (2014) [73], high humidity is known to increase the mass transfer of water vapor in the air, which leads to complex heat and mass transfer problems. As a result, few studies have attempted to use CFD-based modeling to consider both of these conditions to improve reproducibility and explanatory power in the real world.

Secondly, this study goes beyond the micro-spatial scope to consider the meso-spatial scope. The domain of this study is about 5 km × 5 km, and we modeled all the terrain within this range. Of course, for the 1 km × 1 km area including the target site, which is the main analysis area, we additionally modeled the detailed aspects of the buildings. However, this study area has the largest domain and research scope among studies that develop models to describe microclimate based on CFD. At the same time, we modeled specific urban and architectural forms to avail ourselves of the advantages of CFD models. Within this scope, we modeled and studied a meso-scale space that includes the center of the target site, the AWS along the Han River, which is 1.6 km away from the center of the target area, and the AWS on top of the Korea Meteorological Administration building, 2.4 km away from it. In other words, this study is unique in that it attempts to consider climate effects originating from a larger spatial scope while also including the effects of extremely small-scale urban and architectural features.

Third, this study is differentiated by the process and methodology of its model development. More specifically, in order to eliminate the possibility of model overfitting, this study attempts to make a methodological distinction by separating model development and model validation by utilizing data from two AWSs provided by the Korea Meteorological Administration that are sited adjacent to the target site. More specifically, we empirically developed the physical model by iterating the simulation using one set of AWS measurement data. In the

next step, the model was fed with data from the other AWS site, and simulations were performed to compare the results with the actual data measured by the researchers at five locations within the target area. This verified the accuracy and explanatory power of the model, and we attempted to interpret the microclimatic characteristics of the five different urban spaces. This two-step process of model development and validation controls overfitting of the model and attempts to differentiate the methodology in a way that few studies have attempted to verify the explanatory power of the model.

Through this, this study aims to develop a model that can thoroughly predict and explain the impact of urban design plans on reducing or increasing urban heat islands in daytime during heat waves in mid-latitude regions by reviewing urban design plans before performing urban development. In particular, it aims to contribute to the expansion of the related field by developing a Star-CCM + -based model that can quickly review various urban design plans with minimal shape distortion.

Furthermore, this study was conducted in a densely populated residential area with a high degree of urban-architectural complexity that is subject to a highly complex computational environment of high temperature and high humidity as it transitions to a subtropical climate. In particular, the process of developing improved models based on CFD models to describe and predict the behavior of the microclimate in such highly complex environments is one of the challenges facing CFD research on large spaces. Notably, the total domain is 5 km × 5 km, and the detailed modeling of the urban-architectural form, including topography, indicates that the model studied in this research is quite complex. The results of this study are expected to contribute to the methodological development of CFD research in the field of reproducing an external environment with high temperatures and humidity while targeting complex urban and architectural shapes over a large area. In addition, this study applied a two-step methodology to develop a descriptive model of microclimate by separating training data and validation data to develop the model. This attempt at model development and validation methodology is a new step toward providing a descriptive model of microclimate, and we believe that it is a clear contribution to the methodology for creating a descriptive model of microclimate.

2. Methodology

2.1. Study area and local climate

The Noryangjin neighborhood located in Dongjak-gu, Seoul was selected as the study area is. It is located at 37°30'36.30" north latitude and 126°56'00.57" east longitude. The area of the target site is approximately 45,000 m², running 370 m in a north-south direction and 170 m east-west. The Dongjak-gu Office is located in the northwest of the target site, and Yeonghwa Elementary School, Yeongdeungpo Middle School, and Yeongdeungpo High School are located in the southwest. The general location of the subject site and the main areas of interest, including the subject site, are shown in Fig. A.1 and Table A.2

The site is an older residential area that has been discussed as a candidate for urban renewal or redevelopment. The southern part of the site has been designated as Noryangjin Redevelopment Area 6. The site maintains a significant density of low-to mid-rise buildings, which is a characteristic of older residential areas. In addition, due to the high floor-to-ceiling ratio, artificial materials such as concrete and asphalt make up the majority of the total covering. Street trees, parks, and green spaces are virtually absent. This site was chosen for the study because it has all the characteristics of an area with a significant thermal comfort deficit during the hottest summer months. These are a high percentage of artificial cover (over 95 %), and an area occupied by low- and medium-rise buildings reaching 60 %, the maximum limit prescribed by law. This site has unfavorable wind flow and absorbs or reflects most of the insolation, creating a poor thermal environment in the hot summer

Table 1

Previous studies and methodologies by spatial scope of research.

No.	Author (year)		City, Country	Spatial Type	Detailed Spatial Scope	Simulation Methodologies
1	Mayer et al. (2009)	[5]	Freiburg, Germany	spot	2 spots(15m apart)	–
2	Lee and Jim (2019)	[6]	Hong Kong, China	spot	2 spots(30m apart)	–
3	Hsieh et al. (2018)	[7]	Nanjing, China	spot	2 spots(30m apart)	Energy Plus
4	Brozovsky et al. (2022)	[8]	Trondheim, Norway	building	1 building	Fluent + IDA ICE
5	Salata et al. (2016)	[9]	Rome, Italy	building	1 building	Envi-met
6	Nugroho et al. (2022)	[10]	Bandung, Indonesia	building	4 buildings	Fluent
7	Allegrini and Carmeliet (2018)	[11]	Zurich, Switzerland	building	5~9 buildings	OpenFOAM + CitySim
8	Toparlar et al. (2018)	[12]	Antwerp, Belgium	building	365 buildings	Fluent + Energy Plus
9	Lee and Jeong (2018)	[13]	Seoul, Korea	building	22,439 buildings	Envi-met
10	Abuelata (2020)	[14]	Cairo, Egypt	street	1 street(18m)	ENVI-met
11	Abdollahzadeh and Biloria (2022)	[15]	Sydney, Australia	street	1 street(20m, fictitious)	Grsaahopper + Ladybug, Honeybee
12	Zheng et al. (2023)	[16]	6 cities in China	street	6 streets(100m each, fictitious)	BEM + UCM
13	Yamaoka et al. (2008)	[17]	Osaka, Japan	street	2 streets(3 km)	Simulation
14	Qaid et al. (2018)	[18]	Putrajaya, Malaysia	street	1 street(4 km)	Envi-met
15	Li et al. (2021)	[19]	Guangdong, China	region	1 region(126m ² , fictitious)	ENVI-met + TRNYS
16	Ibrahim et al. (2021)	[20]	Cairo, Egypt	region	1 region(651m ² , fictitious)	ENVI-met, EnergyPlus, grasshopper+(Ladybug, Dragonfly, Honeybee)
17	Ahmadian et al. (2021)	[21]	London, UK	region	4 regions(3,600m ² , fictitious)	CitySim
18	Yuan and Ng (2012)	[22]	Hongkong, China	region	3 regions(20,000m ²)	Fluent
19	Zhang and Gao (2021)	[23]	Nanjing, China	region	8 regions(3,519~24,480m ²)	ENVI-met + Energy Plus
20	Nouri and Costa (2017)	[24]	Rossi, Portugal	region	1 region(30,000m ²)	–
21	Maragkogiannis et al. (2014)	[25]	Crete, Greece	region	1 region(34,008m ²)	Phoenix
22	Hsieh et al. (2010)	[26]	Tokyo, Japan	region	1 region(41,760m ²)	Star CCM+/CD
23	Ma and Chen (2020)	[27]	Tianjins, China	region	8 regions(43,200m ² , fictitious)	ENVI-met
24	Fei et al. (2022)	[28]	Tianjin, China	region	1 region(57,600m ²)	ENVI-met
25	You and Ding (2021)	[29]	fictitious	region	1 region(63,504m ² , fictitious)	Star CCM+/CD
26	Battisti et al. (2018)	[30]	Sapienza University, Italy	region	1 region(84,000m ²)	ENVI-met
27	Wang et al. (2016)	[31]	Toronto, Canada	region	3 regions(90,000m ²)	Envi-met
28	Lee et al. (2014)	[32]	Asan, Korea	region	1 region(112,700m ²)	Star CCM+/CD
29	Huang et al. (2005)	[33]	Shinjuku, Japan	region	1 region(125,663m ²)	Star CCM+/CD
30	Detommaso et al. (2021)	[34]	Catania, Italy	region	1 region(129,600m ²)	Envi-met
31	Toparlar et al. (2018)	[35]	Antwerp, Belgium	region	1 region(137,500m ²)	Fluent
32	Gros et al. (2014)	[36]	Nantes, France	region	1 region(140,000m ²)	Envi-BatE model
33	Chen et al. (2009)	[37]	Otemachi & Kyobashi, Japan	region	2 regions (72,770m ² ~177,531m ²)	STAR-CCM+/CD
34	Shui et al. (2018)	[38]	Harbin, China	region	1 region(0.1 km ² , fictitious)	Fluent
35	Müller et al. (2014)	[39]	Oberhausen, Germany	region	1 region(0.2 km ² , fictitious)	ENVI-met
36	Morakinyo et al. (2017)	[40]	Hongkong, China	region	1 region(227,960m ²)	Envi-met
37	Brozovsky et al. (2021a)	[41]	Trondheim, Norway	region	1 region(0.23 km ²)	Envi-met
38	Antoniou et al. (2019)	[42]	Nicosia, Cyprus	region	1 region(0.247 km ²)	Fluent
39	Tominaga et al. (2015)	[43]	Kanagawa, Japan	region	1 region(0.25 km ²)	Fluent
40	Brozovsky et al. (2021b)	[44]	Trondheim, Norway	region	1 region(0.26 km ²)	Fluent
41	Bouyer et al. (2011)	[45]	8 cities in France	region	1 region(0.34 km ² , fictitious)	Fluent
42	Ma et al. (2012)	[46]	Shenzhen, China	region	1 region(0.38 km ²)	ENVI-met
43	Gromke et al. (2015)	[47]	Arnhem, Netherlands	region	1 region (0.54 km ²)	Fluent
44	Toparlar et al. (2015)	[4]	Rotterdam, Netherlands	region	1 region(589,500m ²)	Fluent
45	He et al. (2019)	[48]	Singapore	region	1 region(0.64 km ² , fictitious)	STAR-CCM+/CD
46	Liu et al. (2022)	[49]	Dujiangyan, China	region	9 regions (467,500m ² ~667,500m ²)	Fluent
47	Wise et al. (2018)	[50]	Singapore	region	1 region(1 km ²)	Star CCM+/CD
48	Hsieh et al. (2016)	[51]	Tainan, Taiwan	region	1 region(1 km ²)	Wind perfect
49	Wong et al. (2021)	[52]	Singapore	region	1 region(1.4 km ²)	OpenFOAM + CitySim
50	Wang et al. (2018)	[53]	Hongkong, China	region	1 region(1.44 km ²)	PALM
51	Grifoni et al. (2022)	[54]	Ascoli Piceno, Italy	region	1 region(1.45 km ²)	ENVI-met
52	Fintikakis et al. (2011)	[55]	Tirana, Albania	region	1 region(2.4 km ²)	Phoenix + TRNYS
53	Ketterer and Matzarakis (2015)	[56]	Stuttgart, Germany	region	1 region(6.25 km ²)	ENVI-met
54	Wang et al. (2020)	[57]	Beijing, China	region	1 region(9 km ²)	PALM
55	Javanroodi et al. (2019)	[58]	Teheran, Iran	region	22 regions(14~60 km ²)	Grsaahopper + Ladybug, Energy Plus
56	Priyadarisini et al. (2008)	[59]	Singapore	region	1 region(728 km ²)	CFX
57	Alonso et al. (2020)	[60]	Lyon, France	building	1 building	ArcGIS + Random Forest
58	Zhong et al. (2023)	[61]	Shenzhen, China	region	1 region (300m, fictitious)	OpenFOAM, Grasshopper + CNN
59	Kim et al. (2023)	[62]	Daegu, South Korea	region	1 region(12,000m ²)	STAR CCM+ + GBRT
60	Javanroodi et al. (2022)	[63]	Stockholm, Sweden	region	1 region (25,024m ²)	CFD solver + MLP, DNN
61	Tariku et al. (2023)	[64]	Vancouver, Canada	region	1 region(36,808m ²)	ENVI-met + ANN
62	Adkins et al. (2023)	[65]	Florida, USA	region	1 region(160,000m ²)	Fluent + RNN
63	Mortezazadeh et al. (2022)	[66]	Montreal, Canada	region	1 region (810,000m ²)	CityFFD + Random Forest
64	Alaoui et al. (2023)	[67]	Agadir, Morocco	region	1 region (Total city)	Fluent + ANN, SVM

months. These thermal environmental features were chosen for this study because the summer temperatures of interest in this study can be quite extreme depending on how the simulation model is set up.

2.2. Measurement

The meteorological measurements in this study were taken at five locations within the target site, with measurements taken at a distance of approximately 2 m from the rooftop of the buildings. The measurements were taken over a period of 72 h, from 15:00 on August 9, 2019, to 14:00 on August 12, 2019. Comparisons between the measurements and the simulation were performed using data from 6:00 a.m. to 8:00 p.m. on August 10, which is the middle of the period. The study period fell during a heat wave, with a high temperature of 36.6 °C on August 10, the second hottest day of August 2019, and a high temperature of 36 °C on August 11, the third hottest day of the month. A heat warning was issued by the South Korean government during this period.

The five sites were selected because they were located at the same distance from the center of the target area, the buildings were not particularly tall (around 5–6 m), and the rooftops were available for use. However, at one site (measurement point 4), we were able to obtain the

cooperation of the owner of a vacant lot and the weather meter was installed at a pedestrian level of 2 m above the street. In addition, data loggers were installed on each of the five devices to collect data simultaneously. The measuring instruments used for the meteorological measurements were type-approved by the Korea Meteorological Administration. Their specifications are shown in Table A.1. The study site is a high-density residential area with virtually no public space. In fact, there are no street trees along the throughfares, and there is no distinction between sidewalks and driveways. Most of the buildings have a spacing of four to 6 m between them, which is typical of older residential areas in South Korea. In such a situation, it was impractical to install a weather meter on the roadside for more than 72 h, so the measurements were performed on the rooftops of the lowest buildings available. However, in the case of measurement point 4, there was a privately owned public space and we were able to install the meter at a pedestrian level of 2 m above the street with the consent of the owner. The parameters necessary to define urban form [74–78] of the measurement point are as shown in Table A.2.

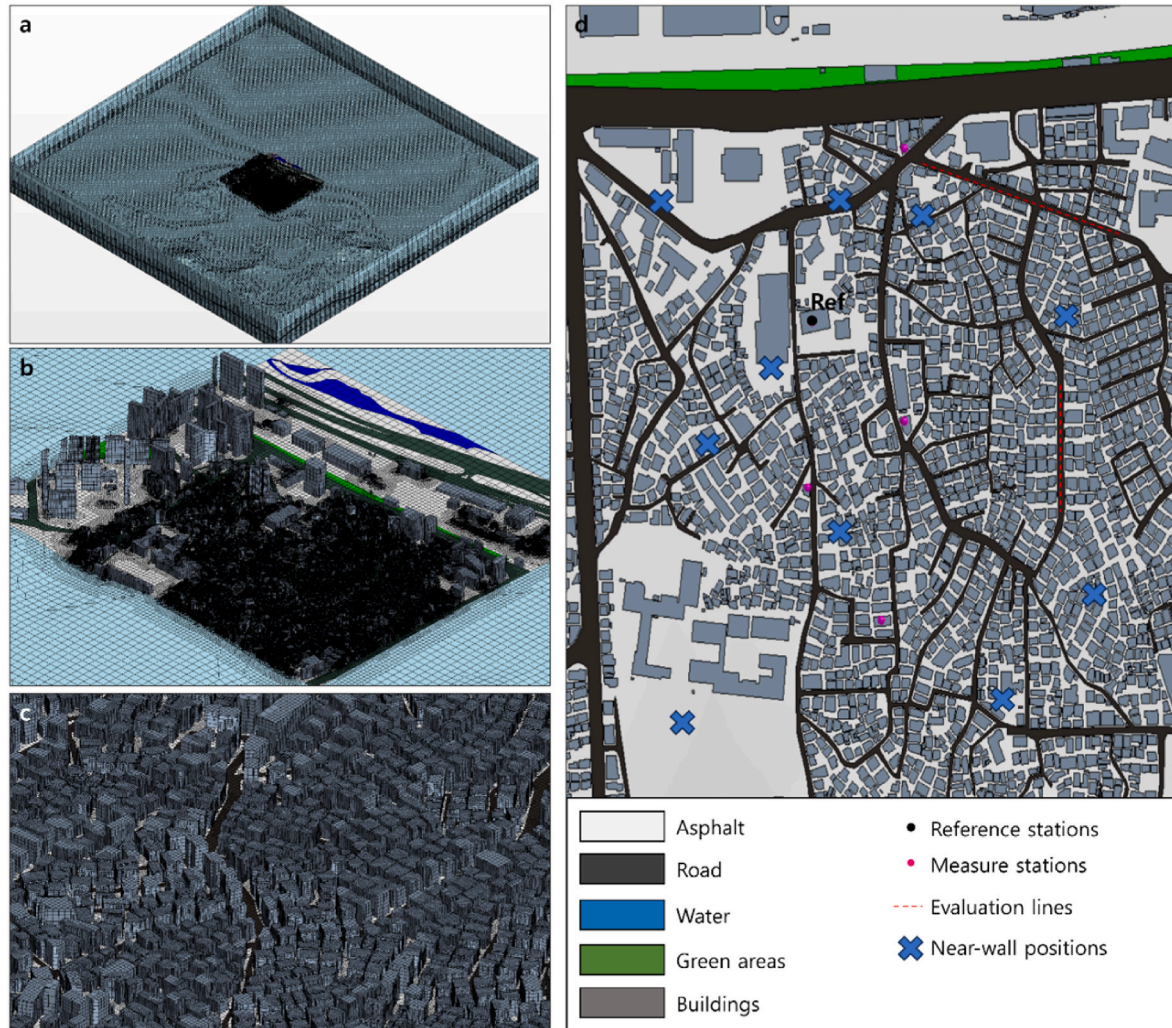


Fig. 1. Computational domain (5060 × 5060 × 515 m) with 13,994,426 cells (a) Full view from the southeast; (b) View of the region of interest with different surface types: Asphalt (light grey), road (black), water (blue), green areas (green), and buildings (dark grey); (c) Close-up view of the residential area; (d) Surface categories, position of the measuring station, the evaluation lines and near-wall position for the grid-convergence analysis. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.3. Computation domain

The computational domain of this study is the Noryangjin area of Seoul (1 km² area). To meet the CFD Best Practice Guidelines by Franke [79] and Tominaga [80], the domain was extended by 2 km (twice the length of the domain of interest) to the east, west, north, and south of the building of interest. The height of the tallest building in the domain is about 90 m, and all ground surfaces are considered to be realistic. The sky was modeled up to 350 m above the ground to form a closed system. Up to 120 m below the ground was modeled to consider subterranean heat storage. The entire domain, including the sky and the ground, has the shape of a hexahedron, as shown in Fig. 1. Since the wind direction changes with time, the side boundaries are both inlets and outlets, which are applied differently depending on the boundary conditions. The resulting domain is 5060 × 5060 × 515 m in size.

The domain was constructed using automated mesh in STAR-CCM+ version 2021.3 and was divided into 13,994,426 trimmed cells (see Fig. A.2). The grid is based on a grid convergence test, which compares three grids of different resolutions to observe their impact on the solution. For this purpose, the wind speed ratios at various locations within the region of interest were compared for a coarse mesh of 6,423,289 cells, a medium mesh of 13,994,426 cells, and a fine mesh of 22,505,185 cells. These locations include the five weather stations, including one reference station, two horizontal lines at a height of 3 m, and a randomly selected near wall position. The main difference between the three grids is the size of the surface mesh within the region of interest. The differences between each of the grids applied to the buildings were compared based on indicators, such as average cell area and maximum cell area, which are shown in Table 2. For all grids, simulations were performed using steady-state Reynolds Averaged Navier Stokes (RANS) with a wind speed of 3 m/s at 10 m above street level in an assumed westerly direction, using the same inlet conditions as described in Section 2.3. All three grids were generated according to BPG [79,80].

The cell expansion rate of 1.2 in the final selected reference meshes ensures a smooth transition in cell size between the high resolution in the region of interest and the coarse mesh in the rest of the domain. The smallest resulting cell in the domain has a volume of 15.82 cm³, while the largest cell has a volume of 9.47 × 10⁵ m³. As recommended by the BPG [79,80], the evaluation height (in this case, 1.75 m, since pedestrian-level climate variables are important) corresponds to the third cell above the near wall cell. The prism layer ensures vertical gridlines on the building walls and the ground surface of the study area.

2.4. Boundary conditions

Since the most important factor affecting the temperature change in the region of interest is insolation, the solar load technique provided by the radiation model of STAR-CCM+ was used. The necessary factors for this method are information about the location and direct/diffuse solar flux, which were applied by entering insolation data for the same date and time in the region as provided by the Korea Meteorological Administration.

For low-level obstructions on buildings and other surfaces, a roughness parameter was used to account for surface roughness factors

in unmodeled areas. STAR-CCM+ does not allow direct input of the aerodynamic roughness length z_0 provided by Wieringa [81], but a roughness height r is required. In this study, a roughness height is applied for each boundary, with 0.25 m for the green side and 0.01 m for the others. Fluid-wall interactions were treated with the standard wall functions by Launder and Spalding [82].

To account for the effect of thermal mass, the ground down to 120 m below the surface was modeled, as mentioned in Section 2.3, and the surface thickness was additionally modeled using the shell region on STAR-CCM+ for all boundaries including buildings in the area of interest as shown in Table B.1. In addition, since the space is large, a large grid was constructed to address practical issues. To numerically interpolate the dissipated convective and radiation heat transfer terms, additional convective heat flux and long wave radiation heat flux were applied, which will be described in detail in Section 2.5. For the sake of simplicity, the wall and roof structures were not distinguished except for the thickness of the shell region, and the same settings were applied to all buildings. All buildings are assumed to be filled with air with an initial temperature of 23 °C, and then the inside temperature is set to change based on the simulation. Water boundary, which was set as a river, is not modeled as a shell region, with the convection condition and wall species source set in the boundary itself to numerically model the decrease in the temperature rise of the surface due to water and the increase in humidity due to evaporation. For the green boundary, which is represented by parks and green areas, the surface of the grass layer is modeled as a shell region, and the heat flux for evapotranspiration of the grass is additionally applied (see Section 2.5). Table B.2. shows the radiation and thermal properties of the boundaries used in the model.

Since the direction of the velocity keeps changing with time in the simulation, the boundaries of all four side domains are set to velocity inlet and applied as velocity vector components. Consequently, the inlet and outlet change in real time according to the wind direction. The air velocity $U(z)$ [m/s] (eq. (1)) at the inlet is used in the form of a logarithmic profile, which is calculated and applied as a height-dependent velocity profile based on the velocity at a height of 10 m above sea level. In eq. (1), u_{Ref} is the velocity at the reference height (10 m) and z is the height. To input the wind direction and wind speed at a height of 10 m, which is the reference height of this research model, the hourly wind direction and wind speed data provided by Automated Weather Stations (AWS) operated by the Korea Meteorological Administration was utilized, just like solar heat flux. The wind direction and wind speed data were converted into direction and speed components. The measured data from the AWS installed on the Korea Meteorological Administration building was used as input for the simulation for model development, and the measured data from the AWS installed along the Han River was used as input for the simulation for model validation. Specifically, we first developed the model using the measured data from the AWS located on the Korea Meteorological Administration building. Next, we used it to empirically verify various parameters such as inflow turbulence intensity and turbulence dissipation rate values. However, after performing simulations with the model so developed, it was determined that there was a high possibility that the model would be overfitted if verification and interpretation were performed with the data measured at five locations in the target area. Therefore, to verify and interpret the developed model, new official data provided by the AWS installed along

Table 2
Surface grid sizing of the three investigated computational grids in the grid-convergence analysis.

	Coarse Mesh (6.4 M cells)	Basic Mesh (14.0 M cells)	Fine Mesh (22.5 M cells)
Average Cell Area [m ²]	4.47	1.63	1.18
Maximum Cell Area [m ²]	17.49	16.15	9.09
Surface Growth Rate [–]	1.4	1.3	1.2
Number of Prism Layers [EA]	2	2	4
Thickness of First Prism Layer [m]	0.25	0.123	0.0615

the Han River were put into the simulation model, and simulations were performed again. Verification and interpretation were performed by comparing the simulation results with the data measured at five locations in the target area using the researchers' own equipment. In addition, since radiation is not an influential factor, both emissivity and reflectivity were set to 0 to allow all solar loads to be transmitted.

Since the turbulence generation in the atmospheric boundary layer is stronger than usual, the values for the inflow turbulence intensity and turbulent dissipation rate at the inlet should be changed to larger values than those provided by STAR-CCM+ [83]. The corresponding equations are eq. (2) and eq. (3). In eq. (2), the turbulent kinetic energy, k [m^2/s^2], is expressed as the square of multiplication of the reference velocity and the dimensionless number of turbulent intensity, TI ($=0.1$), and in eq. (3), the turbulence dissipation rate, ε [m^2/s^3], is calculated as a combination of the von Karman constant ($=0.42$) and the wall distance d_w , which is an internal function of STAR-CCM+. Eq. (4), which is a formula for u_τ used in eq. (3), is calculated from the von Karman constant, the reference velocity, the reference height, and the height of the floor surface. This was applied to the turbulence term in the lateral domain. This, combined with the simulation domain extending 2 km around the region of interest, leads to a realistic atmospheric flow.

Since the top of the domain is not a real plane but a virtual boundary applied to realize a closed system, the shear condition is set to slip, and both emissivity and reflectivity are set to 0 to allow all solar load to be transmitted, just like the side domain.

$$U(z) = u_{\text{Ref}} \left(\frac{z}{10} \right)^{0.33} \quad (1)$$

$$k = (u_{\text{Ref}} TI)^2 \quad (2)$$

$$\varepsilon = \frac{1}{\kappa} \frac{u_\tau^3}{d_w} \quad (3)$$

$$u_\tau = \frac{\kappa u_{\text{Ref}}}{\log \frac{z_{\text{Ref}} + z_0}{z_0}} \quad (4)$$

2.5. Composition of models

The configuration of the model in this study is shown in Table 3 below.

The model was developed using values from the Korea Meteorological Administration AWS installed on the rooftop of the Korea Meteorological Administration building (one of the 25 AWSs installed and operated by the Korea Meteorological Administration in Seoul) as reference values. The final model is used to perform a point-by-point analysis in Chapter 4. In other words, the final model, Model 5, is used to perform additional simulations by entering the measurements of the Han River AWS, which is a different reference value than the one used for the model comparison analysis. The Han River AWS is located in Yeouido at a distance to the target site similar to the Korea Meteorological Administration AWS. Through this process, the suitability of the model proposed in this study was verified. The reference values for creating the model and the reference values for analyzing the final model were different to prevent the model from overfitting.

2.6. Other computational settings

The CFD simulations are governed by the three-dimensional unsteady Reynolds-averaged Navier-Stokes (URANS) equations that apply a realizable k - ε turbulence model [84]. This turbulence model has been successfully used in other validated CFD studies on urban environment simulations [85–89] and is recommended over Franke et al. [79]'s standard k - ε model. The multi-component technique provided by STAR-CCM+ was used to consider humidity, and the version suggested by Buck [90] was applied as an empirical formula for the saturated vapor pressure required to change the composition of air and vapor to the form of humidity we know. Eq. (5) shows that the saturated vapor pressure p_{sat} is expressed as a function of temperature T [$^\circ\text{C}$].

$$p_{\text{sat}} = 611.21 e^{\left(\frac{18.678 - \frac{T}{234.5}}{257.14 + T} \right) T} \quad (5)$$

Land has a temperature gradient due to the difference in temperature between the surface and the areas below, up to about 15 m. It remains at about 16°C below that. To reflect this, a table was used for the setting of the initial condition of the land where the temperature gradient was set up from the surface to 15 m, and it is fixed at 16°C at greater depths to more realistically reflect the thermal mass of the land. The temperature distribution graph according to soil core applied to this model is shown in Fig. B.1.

As mentioned in Section 2.4, since the domain is too large and there are many buildings, the number of grids is insufficient compared to the BPG. To compensate for the inaccuracy of the solution, convective heat flux, long wave radiation heat flux, and evaporation heat flux were additionally applied to each boundary to numerically interpolate the dissipative convective and radiative heat transfer terms.

Convective heat flux can be described by a heat transfer equation such as eq. (6). According to this equation, h can be expressed as the convective heat transfer coefficient [$\text{W}/\text{m}^2\cdot\text{K}$]. This h utilized the McAdams formula [91] that applies wind speed (eq. (7)).

$$H = h(T_{2m} - T_s) \quad (6)$$

$$\begin{aligned} h &= 5.7 + 3.8\mu \text{ for } \mu < 4.88 \text{ m/s} \\ h &= 7.2\mu^{0.78} \text{ for } \mu \geq 4.88 \text{ m/s} \end{aligned} \quad (7)$$

Long-wave radiation heat flux is a formulation of the heat flux resulting from the temperature difference between the Earth's surface and the sky. The daily average net flux for long-wave radiation is given by eq. (8). This formula is expressed as the difference between the temperature of the sky and the temperature of the ground. $\alpha = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2\cdot\text{K}^4)$ is the Stefan-Boltzmann constant and ε is the emissivity of the surface of the ground. Also, C_{LW} is a constant equal to $4.83 \text{ W}/(\text{m}^2\cdot\text{K})$. The second part of eq. (8) is a linearized form for easier modeling. This approximation is the absolute value of the difference between the ground temperature (T_s) and the sky temperature (T_{sky}). The sky temperature (T_{sky}) depends on the air temperature and humidity and changes during the day and night. This sky temperature (T_{sky}) can be expressed by the empirical formula in eq. (9) [92]. T_{dp} is the dew point temperature ($^\circ\text{C}$), and T_{sky} and T_a are expressed in Kelvin. Finally, t is the time measured after midnight, expressed by hours.

Table 3
Model composition.

Boundary	Basic Physical condition	Land Conduction	Longwave Radiation	Convection	Evapo-transpiration
Model 1 (Base)	○				
Model 2	○	○			
Model 3	○	○	○		
Model 4	○	○	○	○	
Model 5	○	○	○	○	○

$$LW = \varepsilon \alpha (T_s^4 - T_{sky}^4) \cong \varepsilon C_{LW} (T_s - T_{sky}) \quad (8)$$

$$T_{sky} = T_a \left[0.711 + 0.0056 T_{dp} + 0.000073 T_{dp}^2 + 0.0013 \bullet \cos(15t) \right]^{1/4} \quad (9)$$

Moisture flow is caused by the partial pressure difference of water vapor in the gaseous state at the surface. This mass flux is associated with the heat flux transfer required for the evaporation of water, and the land surface loses heat as water evaporates [92]. Evaporative heat flux can be defined according to eq. (10), assuming a linear relationship between saturation vapor pressure and temperature. Here, C_{EV} is treated as a constant and has a value of $0.0168 \text{ }^\circ\text{C}/\text{Pa}$ [92], f is the evaporation rate coefficient, which has a value of -0.05 , and RH is the relative humidity of the air ($-$).

$$EV \cong C_{EV} f h [P_{sat} - RH \bullet P_{sat}] \quad (10)$$

In the case of greenery, the Penman-Monteith Equation (eq. (11)) was applied for crop evapotranspiration since the evapotranspiration of grass is different from that of a typical wall. In eq. (11), R_n is net radiation at the crop surface ($\text{MJm}^{-2}\text{day}^{-1}$), G is soil heat flux density ($\text{MJm}^{-2}\text{day}^{-1}$), T is air temperature at 2 m height ($^\circ\text{C}$), u_2 is wind speed at 2 m height (ms^{-1}), e_s is saturation vapor pressure (kPa), e_a is actual vapor pressure (kPa), Δ is the slope vapor pressure curve ($\text{kPa}^\circ\text{C}^{-1}$), and γ is the psychrometric constant ($0.067 \text{ kPa}^\circ\text{C}^{-1}$). The detailed formula for Δ is summarized in eq. (12), where Δ is defined as a function of temperature, and R_n - G is applied as the calculated boundary radiation heat flux.

$$ET_o = \frac{0.408 \Delta (R_n - G) + \gamma \frac{900}{T+273} \mu_2 (e_s - e_a)}{\Delta + \gamma (1 + 0.34 \mu_2)} \quad (11)$$

$$\Delta = \frac{4098 \left[0.6108 \exp \left(\frac{17.27T}{T+237.3} \right) \right]}{(T+237.3)^2} \quad (12)$$

For road, asphalt, and building boundaries, the combined values of convective heat flux, long-wave radiation heat flux, and evaporation heat flux were applied as the total heat flux. For green, the two values of convective heat flux and crop evapotranspiration were applied as the total heat flux to compensate for the numerical deficiency.

3. Results

3.1. Grid-convergence analysis

This section summarizes the results of the grid-convergence analysis. The settings for grid size and computation are described in Section 2.3. Fig. 2 shows that the difference between the basic grid (13,994,426 cells) and the coarse grid (6,423,289 cells) is very large for weather stations, near-wall positions, and extra evaluation lines. On the other hand, the difference between the default grid and the fine grid (22,505,185 cells) is very small, indicating that further refinement of the grid in the default grid would have a negligible effect on the numerical analysis results. This suggests that the default grid is a good compromise between computational cost and accuracy. Therefore, the basic grid is used as the lattice configuration setting for this study.

The technical specifications of the computer utilized to perform the simulations in this study are as follows: The CPU is an Intel(R) Xeon(R) Gold 6248R CPU @ 3.00 GHz (40 cores); the memory is 383 GB, and the GPU is an NVIDIA Quadro RTX 5000. Using this, the solving time for the final model was 21.47 h. We have mentioned this in the text.

3.2. Simulation results

3.2.1. Temperature changes by model

As mentioned earlier, the temperature changes according to each model were compared with the experimental values for the five points (see Fig. 3). As with the full simulation, the changes were observed from 6am to 8pm.

Point 1 (Top) shows that the simulated results are generally greater than the measured results. The difference between the simulated and measured results increases starting at 8:00 a.m., and the difference between the simulated and measured results becomes significant around 1:00 p.m. for the base model and 12:00–12:30 p.m. for the other models. For the base model (Model 1) and for Model 2, the difference between the simulated and measured values is about $4.5\text{--}5 \text{ }^\circ\text{C}$. This difference decreases as the model becomes more sophisticated, but even for the most sophisticated model, Model 5, the difference is relatively large compared to other points, with an instantaneous difference of up to $3.7 \text{ }^\circ\text{C}$ around 12:10 p.m. This difference decreases after 1:30 p.m., and by 5:30 p.m., the simulated and ground truth results are the same. The average difference between the simulation results and the ground truth for Model 5 is about $0.92 \text{ }^\circ\text{C}$, which is greater than the differences at other points.

Point 2 (left) is a point where the simulated and measured results are quite similar, and even at 12:30 p.m., when the difference is the largest,

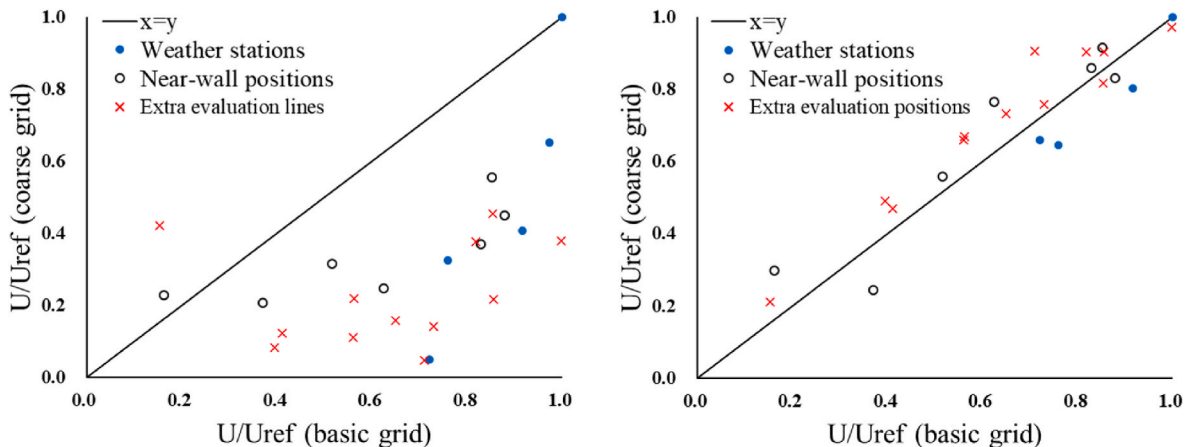


Fig. 2. Comparison of wind speed ratios for the coarse and the basic grid (left) and the fine and the basic grid (right) at the measurement points, evaluation line.

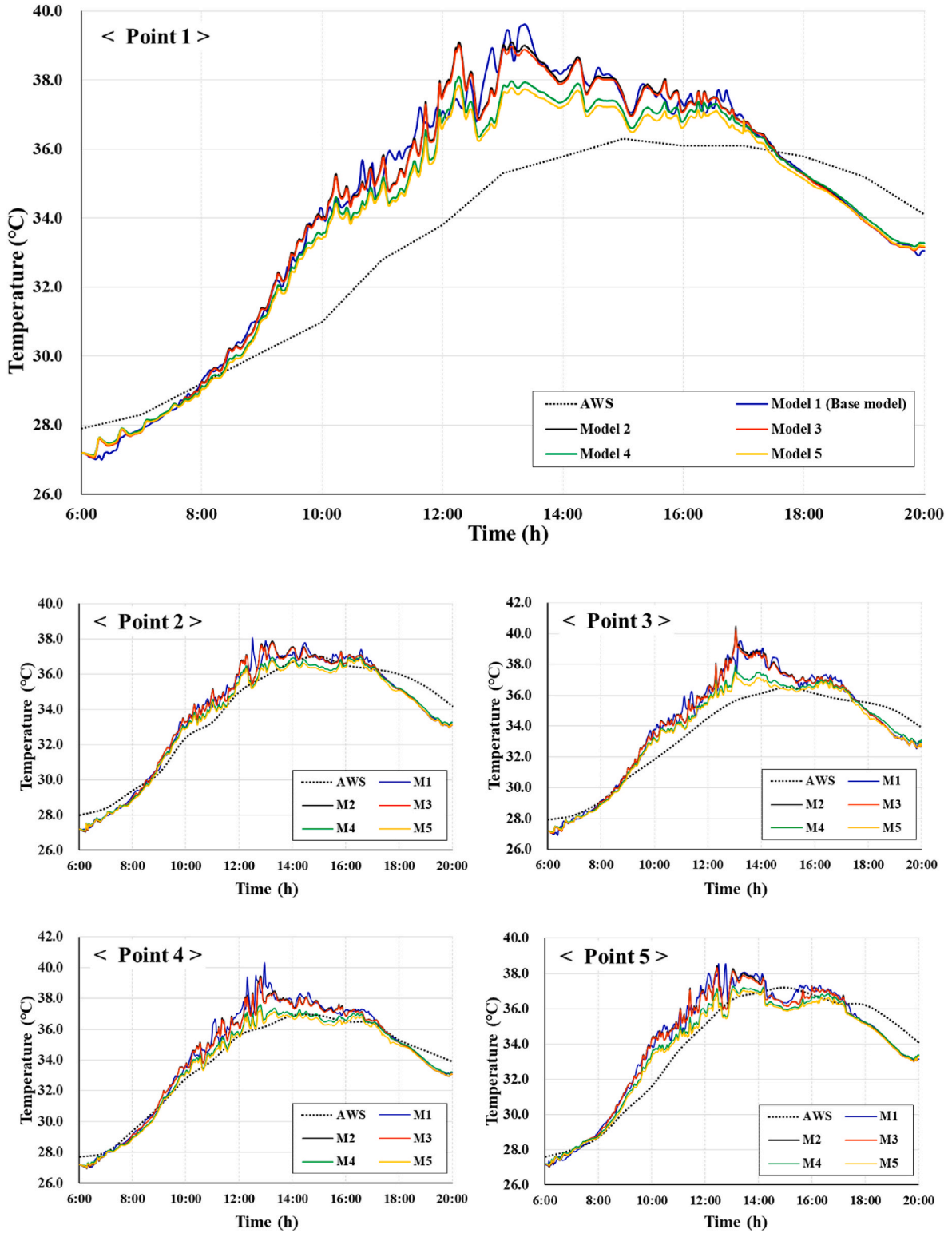


Fig. 3. Changes in temperature for each point by model.

the difference is about 1.8 °C for Model 2 and 1.1 °C for Model 5. In general, Model 5 shows an average difference of 0.03 °C from the actual data, which is virtually indistinguishable considering the temperature sensor's accuracy of ± 0.2 °C. From 2 p.m. to 5 p.m., the values are exactly the same, and from 5 p.m. onwards, the simulated values tend to be up to 1.2 °C lower than the actual values.

At point 3 (right), the difference between simulated and measured results is smaller than at point 2, but the difference is generally similar. In particular, the improvement effect is large as the model is upgraded, and at 12:10 p.m., when the difference is large, the difference between the simulated and measured results is up to 3.6 °C for Models 2 and 3, but the difference improves to 1.2 °C for Model 5. The average difference between Model 5 and the actual measurement is about 0.38 °C, which is outside the error range, but we can observe that the difference is quite small.

At point 4 (left), the difference between the simulated and measured results is smaller for the final model, Model 5, than at point 2. Even at 12:10 p.m., when the temperature difference was the largest, it was only 0.8 °C. On average, the difference between the simulated and measured values was 0.09 °C in the case of Model 5, which is also very small considering the error range of the temperature sensor. In particular, point 4 showed a significant improvement from the base model (Model 1) to Model 5. At 12:10 p.m., the difference between the simulated and measured results of the base model (Model 1) was up to 3.9 °C, while the difference between the simulated and measured results of Model 5 was 0.8 °C, which is an improvement of about 3.1 °C or greater.

Finally, point 5 (right) shows a somewhat different pattern compared to the other points. Especially from 2:20 to 4:00 p.m. and after 5:00 p.m., the simulated temperature is lower than the measured temperature in all models. The largest temperature difference was observed at around 12:20 p.m., where the difference between the simulated and measured results of the base model (Model 1) or Model 2 was up to 3.1 °C. The largest difference when the simulation results were cooler than the actual results was observed at 2:50 p.m., with a difference of 1.1 °C for the final Model 5. On average, the difference between the actual and simulated results is 0.09 °C, which can be considered virtually the same value considering the error range of the temperature sensor.

3.2.2. Comparison of CFD with field measurements

Correlation coefficient, R^2 , mean bias error (MBE), root mean square error (RMSE), were used to compare the CFD simulation results with the actual results. The results of this comparison are shown in the Table 4. In general, the correlation coefficients were high, above 0.9, and the R^2 s were between 0.85 and 0.91. The highest R^2 s were found for Points 4 and 5.

3.2.3. Change in energy for each heat flux

Based on Model 5, the heat flux of each property for land, roads, buildings, and greenery was observed since it allows an indirect explanation of why the energy reaching the ground from the sun is converted into temperature change (see Fig. 4). As with the full simulation, the change from 6:00 a.m. to 8:00 p.m. was observed.

First, the heat flux of the land (left) and the road (right) showed a similar pattern. However, the evapotranspiration value was larger for the land than for the road, and the convection effect was greater for the road. This is because the temperature difference between the upper and

lower portions, which is a condition for convection, is greater at the top of the road.

Next, buildings (left) showed a smaller overall heat flux, while greenery (right) showed a different pattern compared to the other types. The heat flux due to evapotranspiration was large and, conversely, the heat flux due to convection mostly had a positive value because the temperature difference between the upper and lower air was small.

3.3. Verification of the final model

A comparison of the previous five models showed that the final model, Model 5, was the best (see Table 4). Based on this Model 5 we conducted a final verification by replacing the input data from the AWS installed on the building of the Korea Meteorological Administration with the data measured by the AWS installed along the Han River and performing the simulation again. As mentioned earlier, this was to prevent overfitting of the model and to secure universality of interpretation by checking whether the explanatory power of the model is well maintained even if slightly different data is entered to the developed model. Based on the final model, a final verification and interpretation was attempted by comparing the temperature, humidity, and wind speed obtained through simulation at each point.

For temperature, the correlation coefficients ranged from 0.98 to 0.99. With R^2 , they ranged from 0.96 to 0.98. With RMSE, they ranged from 0.79 °C to 1.53 °C. With ER, the range was from 0.63 % to 1.25 %. The R^2 was highest at Point 2 and lowest at Point 4, the RMSE and ER were highest at Point 1, the RMSE was lowest at Point 5, and the ER was lowest at Point 2 (see Fig. 5, Table 5).

Overall, the R^2 , which ranged from 0.85 to 0.91 when AWS measurements were used to build the model, increased to 0.96 to 0.98, indicating that the fitness of the model improved. Overall, the simulation results were generally lower than the ground truth. This is due to lower input temperature values (33.0 °C, 32.3 °C) and higher wind speeds (1.71 m/s, 2.75 m/s) than the AWS values used earlier.

For humidity, the correlation coefficients ranged from 0.98 to 0.99. With R^2 , they ranged from 0.96 to 0.98. With RMSE, they ranged from 2.41 % to 3.38 %. With ER, they ranged from -2.60 % to -1.06 %. The R^2 is highest at Point 2 and lowest at Point 1, while the RMSE and ER are highest at Point 3 and lowest at Point 4 (see Fig. 6, Table 6).

In the case of wind speed, the difference between the simulated and measured values was quite large because 6:00 a.m. was the start of the simulation and the inputs of the reference AWS were not sufficiently simulated. Therefore, unlike with the other factors, the error of wind speed was verified with 6:00 a.m. excluded. The error of wind speed by indicator is as follows: Correlation coefficients ranged from 0.80 to 0.90; R^2 from 0.64 to 0.81; RMSE from 0.14 m/s to 0.24 m/s; and ER from -0.14 % to 0.12 %. The deviation from the previous temperature and humidity data was significant. The R^2 was highest at Point 4 and lowest at Point 1, the RMSE was highest at Point 3, the ER was highest at Point 1, the RMSE was highest at Point 2, and the ER was lowest at Point 4 (see Fig. 7, Table 7).

4. Discussion

Overall, Points 1, 2, and 3 show a high reproducibility between simulation results and ground truth, while Points 4 and 5 show a low reproducibility. Points 1, 2, and 3 are relatively open areas, especially in the panoramic view. There were generally fewer man-made structures blocking the measurement points. In contrast, site 4 is located at the lowest pedestrian height, unlike the other sites, and site 5 was surrounded by large structures. This may have affected the simulation of wind speed and therefore the reproduction of temperature and humidity.

Regarding individual sites, site 1 has a small error in temperature and wind speed, but a somewhat lower correlation for humidity and wind speed. As mentioned earlier, site 1 is in an open field but there is a

Table 4
Comparison between simulated values and actual values by model.

	Corr	R^2	MBE [°C]	RMSE [°C]
M1	0.929	0.864	-0.593	1.325
M2	0.926	0.858	-0.565	1.304
M3	0.928	0.863	-0.534	1.267
M4	0.955	0.913	-0.240	0.892
M5	0.955	0.913	-0.091	0.879

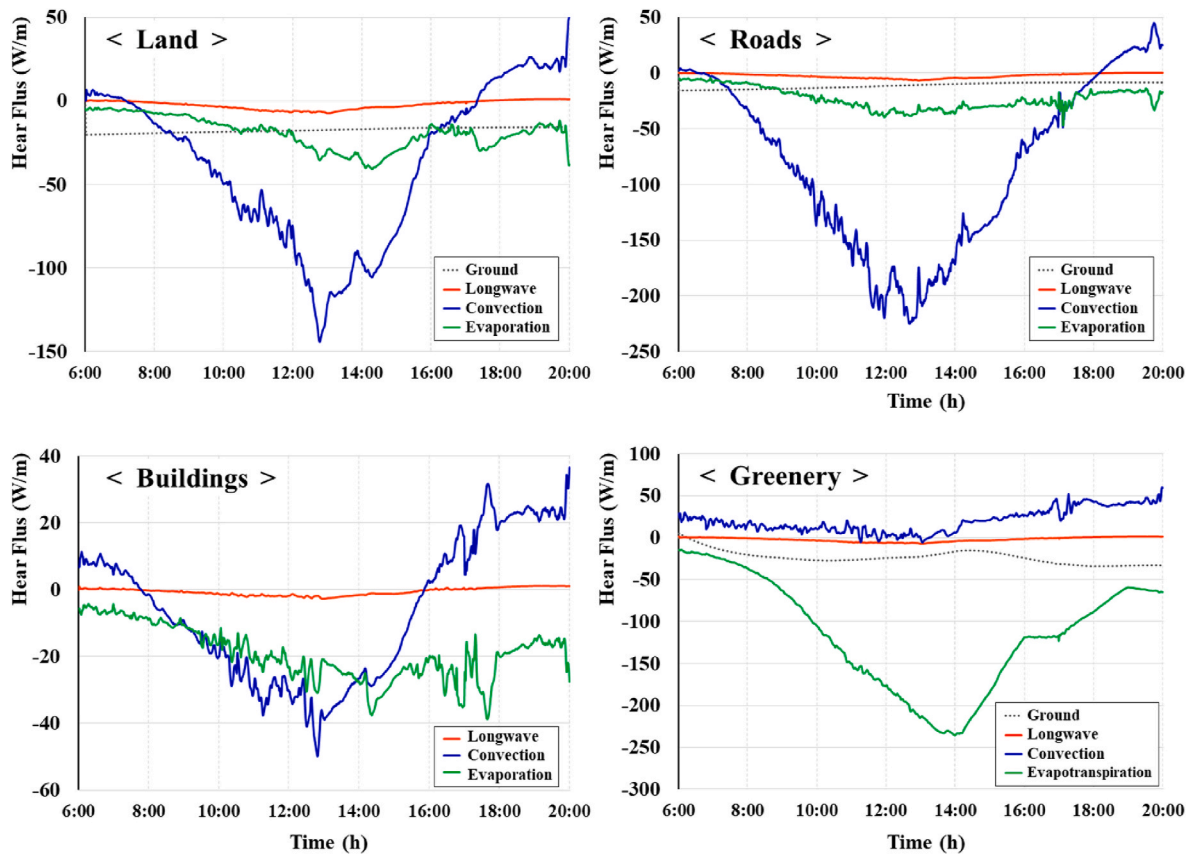


Fig. 4. Changes in heat flux for each property by hour.

large parking lot of about 80 spaces in an adjacent area. Site 1 is actually paved with asphalt, and for this study it was modeled in the simulation as having the same material properties as a road. The heat flux in this part of the parking lot should be reconsidered; it is an open area, so the error in the wind speed simulation is small, but it is believed that there are parts where the temperature and wind speed are unstable, differing from the smooth tendency of the actual measurement. This suggests that the properties of the asphalt comprising the road should be re-examined.

For Points 2 and 3, the trends of the actual and simulated results are quite similar. However, relatively large errors were observed, especially the wind speed at Point 2. The fluctuations of the actual measurements at Point 2 are larger than in the simulation, which is thought to be due to the relatively large trees planted on the rooftop (which is the measurement point). Since the simulation does not take this into account and simplifies the modeling, the simulation results are relatively smoothed, which explains the large error. Nevertheless, the fact that the actual sudden drop in wind speed around 12 o'clock was also simulated is considered an adequate achievement in terms of simulation reproducibility.

Next, Point 4 shows poor reproducibility in terms of both trend and error. Unlike the other points, Point 4 is located at ground level and is surrounded by many small and medium-sized buildings. Therefore, it can be expected that there would be significant difficulties in simulating wind speed. The convection effect, which accounts for the largest heat flux in this study's model, is highly dependent on wind speed, so Point 4 seems to be in a highly unfavorable condition for temperature and humidity reproduction.

Finally, Point 5 is similar to Point 4 in that it is surrounded by a number of buildings, but it shows better reproducibility than Point 4. This is true even though it is relatively less reproducible than the other points due to the relatively large size of the surrounding buildings and the fact that the measurement point is on a rooftop.

The study endeavored to precisely match the points where the actual measurements of temperature, humidity, and wind speed were taken in the field with the points where the values for the simulation results were extracted. In addition to latitude and longitude, height above street level was also accurately matched and compared. As a result, it was found that temperature and humidity were less reproducible at one site (Point 4), which was installed at pedestrian level, compared to the four sites (Points 1, 2, 3, and 5) on relatively open rooftops. However, wind speed was better reproduced at site 4 than at the other sites, indicating that wind speed and turbulence were well simulated. However, in order to reproduce temperature and humidity, there are quite a few factors other than wind, such as the representation of various properties that make up the urban environment, the physical model, and even the accuracy of the modeling, which shows that there is still room for improvement in this research model.

In addition, there were limitations in the simulation of longwave radiation flux and a need for further research. As shown in the study by J. Görtz et al. (2022) [93], longwave radiation flux is expected to be the second-most energetic after convective flux. In this study, a similar pattern of increasing longwave radiation flux was observed from morning to afternoon and decreasing towards evening, but the magnitude was smaller. The reason for this is related to evaporative heat flux

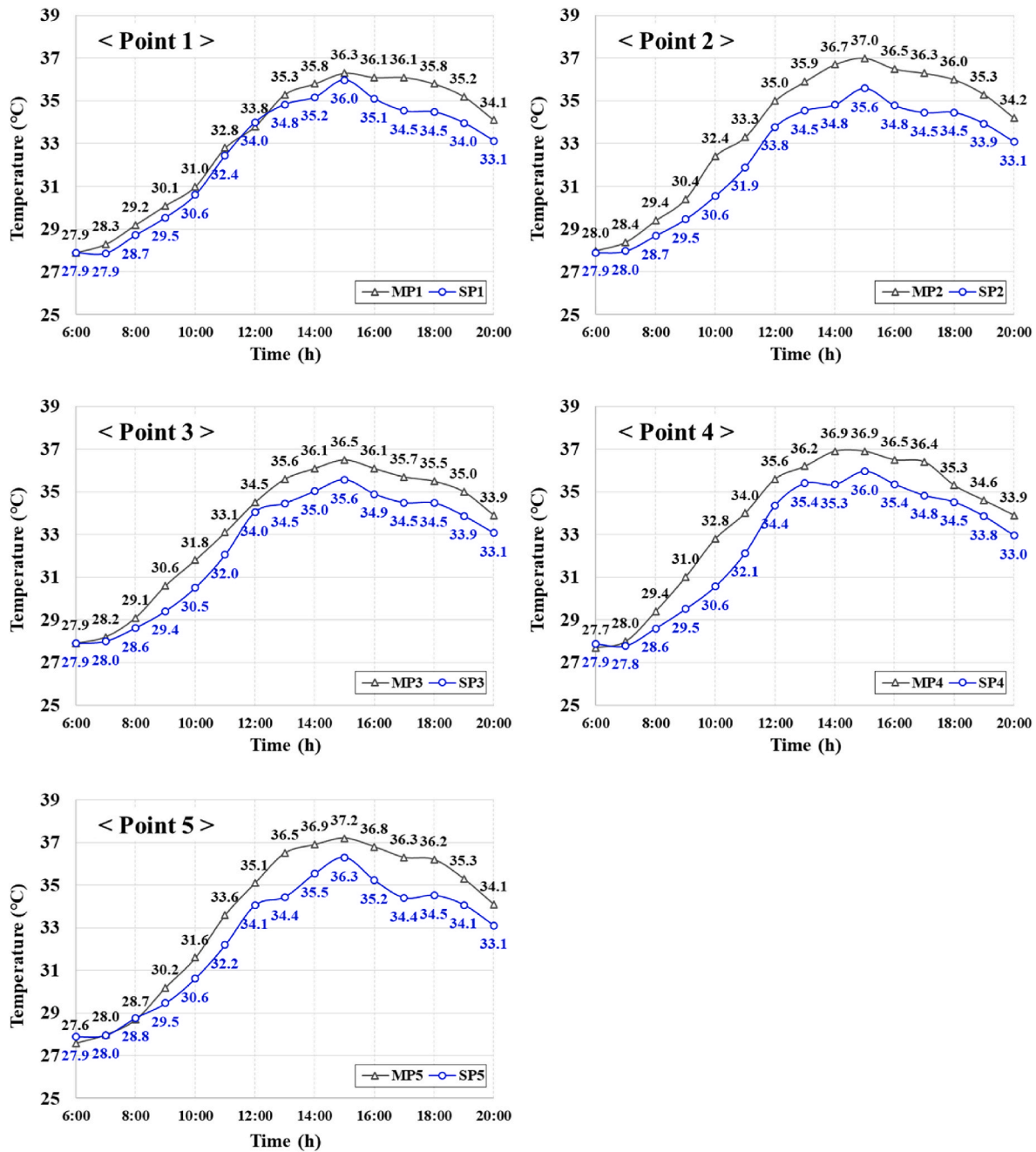


Fig. 5. Comparison of simulated and measured temperatures for each point.

Table 5

Comparison of simulated and measured temperatures for each point by indicator.

	Corr	R ²	RMSE [°C]	ER [%]
P1	0.9892	0.9785	0.7922	0.6353
P2	0.9945	0.9891	1.3558	1.2569
P3	0.9942	0.9885	0.9667	0.8836
P4	0.9829	0.9662	1.2321	1.0767
P5	0.9929	0.9859	1.5354	1.0372

from road surfaces and building surfaces. In this study, various efforts were made to improve the explanatory power of the model in a hot and humid climate. Indeed, as pointed out by Keller Sullivan Oliveira Rocha et al. (2014) [73], CFD results are characterized by a lack of ease in

modeling the various physical effects caused by water vapor. More specifically, high humidity increases the mass transfer of water vapor in the air, which leads in turn to complex heat and mass transfer problems. It is also known that the nonlinear coupling effects between heat transfer and airflow become more pronounced in hot and humid environments, complicating simulations and making accurate modeling difficult. However, in the extremely hot and humid environment of 85 % humidity and 36 °C or higher peak temperatures that took place in this study, we observed that the CFD results are quite unstable. In fact, over the course of the study, we observed that the modeled surface temperatures of roads and buildings increased to more than 80 °C, and the air temperature reached 45 °C, as was the case in the studies mentioned in the introduction. This is not the case in actuality. To address this issue, this study focused on the disturbance of long-wave radiation by water

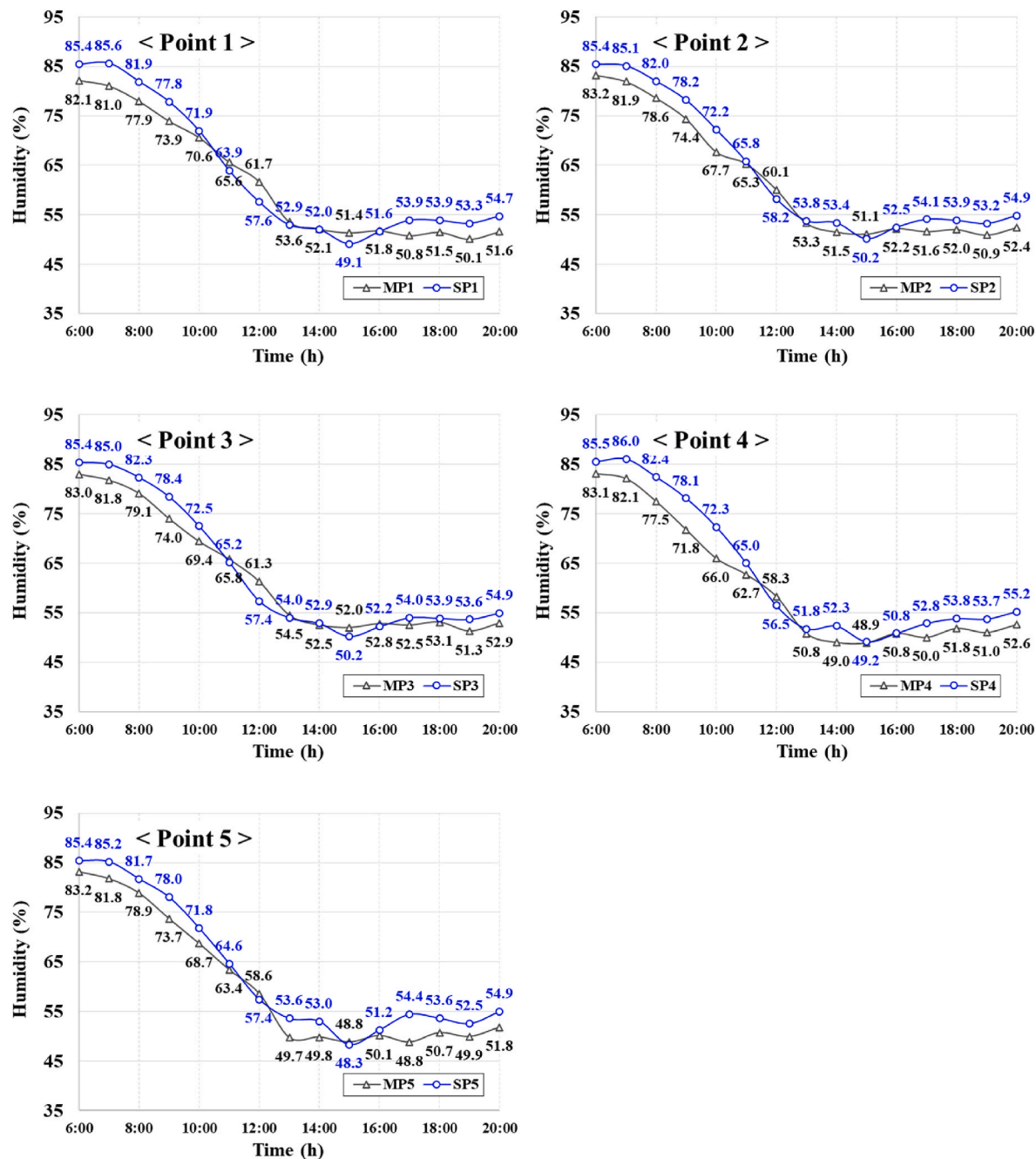


Fig. 6. Comparison of simulated and measured humidity for each point.

Table 6

Comparison of simulated and measured humidity for each point by indicator.

	Corr	R ²	RMSE [%]	ER [%]
P1	0.9835	0.9673	2.8970	−1.3229
P2	0.9929	0.9858	2.4683	−1.7732
P3	0.9897	0.9795	2.4127	−1.0634
P4	0.9900	0.9802	3.3814	−2.6027
P5	0.9916	0.9833	3.0396	−2.5216

vapor, the condensation of water vapor at night, and the effect of the evaporation of water vapor on the surface temperature of buildings and roads. At the micro level, although there are few trees such as street trees in the target area, there are many privately planted trees or shrubs adjacent to the roofs of buildings or their outer walls. We assumed that

their influence could not be overlooked, so we introduced evaporation efficiency into the built environment by referring to the material-specific physical parameters from the studies of Richard A. Anthes et al. (1987) [94], Ichinose et al. (1999) [95], and Yasunobu Ashie & Takaaki Kono (2011) [96], and set the road and building surfaces to dissipate high energy. This allowed us to achieve high explanatory power, as shown in this study, but it also resulted in a situation in which evaporative heat flux appeared from the road and building surfaces, reducing the long-wave radiation flux. This is a limitation of this study and requires further research. Furthermore, a review of models or methods to confirm or improve this phenomenon is needed.

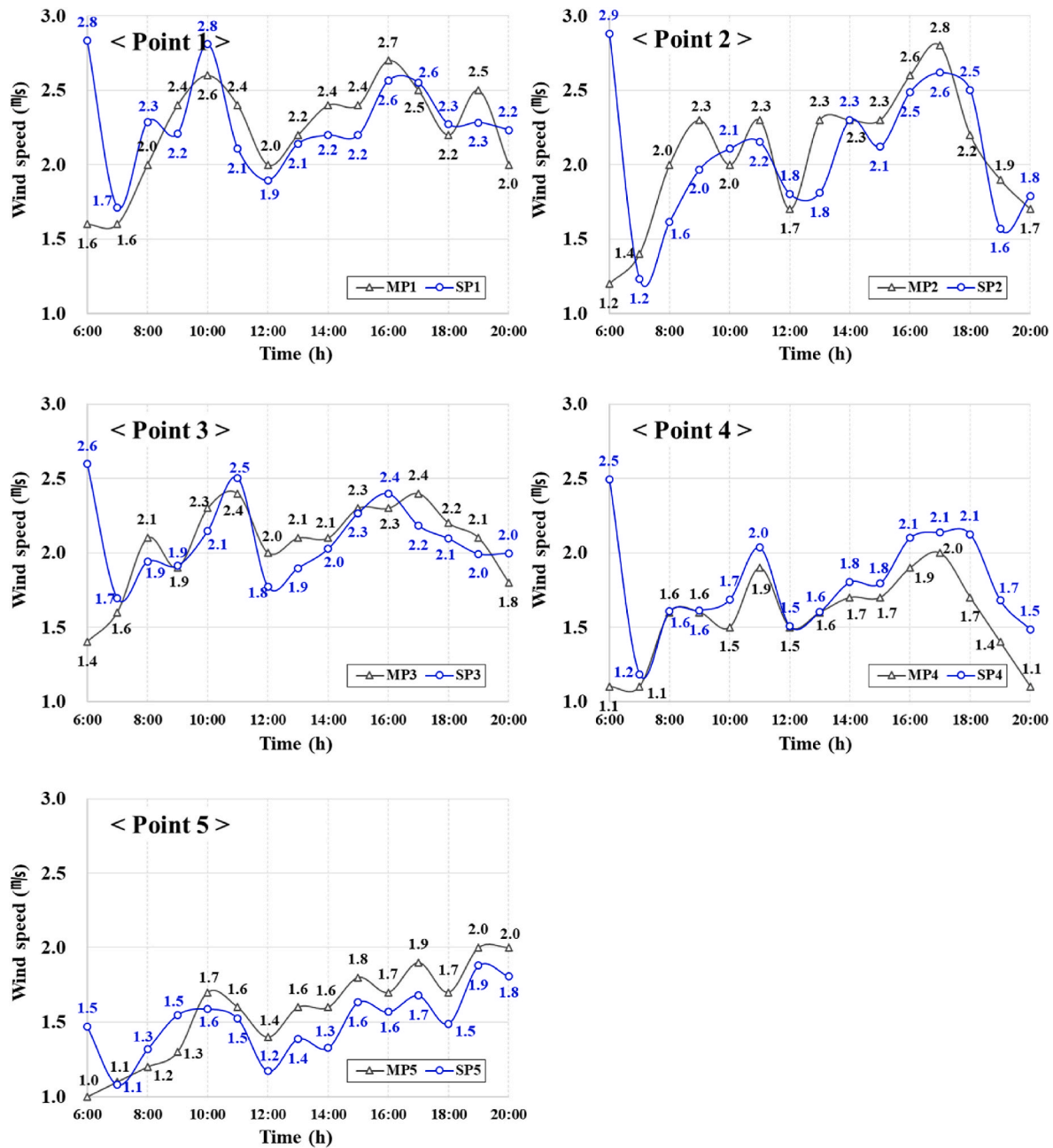


Fig. 7. Comparison of simulated and measured wind speeds for each point.

Table 7

Comparison of simulated and measured wind speeds for each point by indicator.

	Corr	R ²	RMSE [m/s]	ER [%]
P1	0.8047	0.6475	0.1864	0.0319
P2	0.8409	0.7070	0.2472	0.1234
P3	0.8618	0.7427	0.1430	0.0554
P4	0.9007	0.8112	0.1977	-0.1474
P5	0.8300	0.6889	0.1807	0.1145

5. Conclusions

This study aimed to develop a CFD model that reproduces daytime conditions during the hottest and most humid summer months in an older neighborhood in Seoul with high density of low- and mid-rise buildings. The CFD model defined in this study models the buildings,

the man-made and natural coverings, and, to a lesser extent, the vegetation in the target area. The model proposed in this study includes various thermal effects such as shortwave radiation from the sun, longwave radiation due to solar reflection and absorption by various coverings, heat storage and exchange from buildings, heat storage and exchange with the ground and groundwater, evapotranspiration effects from different coverings due to vegetation and atmospheric moisture, and convection effects due to temperature differences and wind speed.

To validate the CFD model proposed in this study, five meteorological stations were installed for a total of 72 h spanning three days. These fixed stations provided measurements of hourly average wind speed, air temperature, and humidity. The validation time selected for this study was from 6:00 a.m. to 8:00 p.m. (totaling 14 h) on the second day of the 72-h period. For the CFD simulation, the input values were not the values of the five fixed weather stations at the target site, but the official values as measured by the AWS maintained by the Korea Meteorological

Administration. There are two AWSs operating near the target site of this study, and the models from the first model (base model) to the fifth model were developed with data from one of them. The final verification was performed by entering the data of the other AWS into the final model that showed the best fit in an effort to prevent the model from overfitting the data. As a result, the CFD model proposed in this study was observed to have a reasonable level of reproduction in the five validation measurements.

Comparative validation with the five weather stations in the field showed that temperature had an R^2 of 0.96–0.98 with an RMSE of 0.79–1.53 and an ER of 0.63–1.25, while humidity had an R^2 of 0.96–0.98 with an RMSE of 2.41–3.38 and an ER of –2.60 to 1.06. For wind speed, the R^2 ranged from 0.64 to 0.81, the RMSE ranged from 0.14 to 0.24, and the ER ranged from –0.14 to 0.12. It is expected that the deviation between points is affected by the spatial characteristics of each point and the surrounding area. From these results, it can be concluded that the temperature, humidity, and wind speed in the low-level atmospheric boundary layer are reproduced with sufficient accuracy by the CFD model proposed in this study.

The novelty of this study can be summarized in three ways. First, the target area has been within a temperate climate region for a considerable time, but due to recent changes in the climate, the temperature and humidity are rising. The climate pattern is gradually shifting to something similar to that of a subtropical climate region. While there have been many studies of urban microclimates and urban heat islands, few have attempted to model the extreme heat and humidity that have been imposed by climate change on otherwise unaffected areas. Second, there have been few attempts to model these unique environments based on CFD models. CFD models are recognized for their accuracy, especially when simulating indoor environments with controlled external environmental variables. Despite the fact that CFD models have the advantage of being able to provide specific interpretations based on highly detailed modeling, they face significant challenges when analyzing large areas. In particular, large spaces, complex urban and architectural geometries, as well as high temperatures and high water vapor content in the atmosphere all increase the complexity of CFD simulations, expanding computational time and reducing accuracy. The novelty and distinction of this study is that it addresses this situation and simulates a large space and high temperature and humidity environment based on CFD. Finally, the use of two sets of officially available data created at different locations to develop and validate the model is a methodological advance that is not commonly adopted by related studies. Although such model development and validation methods have been utilized in various studies, no CFD-based model development and validation studies have adopted them to prevent overfitting of the model and provide more thorough validation of the developed model.

The model established in this study can be applied to cities in a wide range of temperate climates that are experiencing climate change, not just those in the study area. This regional scope includes much of East Asia, which is home to a quarter of the world's population. Therefore, the CFD model presented in this study can be expected to be utilized in

the future to predict microclimate conditions during summer and in heat waves in low- and medium-rise areas of cities with high density. Especially, it is expected that this CFD model can be utilized in the future to predict the microclimate environment during the summer and in periodic heat waves for cities with a high density of mid- and low-rise areas. By utilizing the model developed in this study, it is expected that it will be possible to predict and explain more accurately the environmental impacts from urban development of summer and of heat waves resulting. Utilizing this research model, it would be possible to reduce the health impacts on urban residents, reduce the energy consumption of buildings in such areas, and promote outdoor activities in the external space of urban residents. In other words, it is possible to derive urban design plans that can minimize the social and economic costs of climate change and urban heat islands by controlling the temperature of urban space through artificial interventions.

However, this study is limited by the fact that the results were only evaluated for the daytime hours of 6:00 a.m. to 8:00 p.m. The study needs to be more comprehensively validated for a wider range of urban conditions. In addition, this aspect of the modeling needs further review and research, as the target site is mostly composed of flat roofs and has no vegetation except for a few green areas. Further research is needed on modeling and the consideration of turbulence to more accurately simulate wind, including on the elevation differences between points, the effects of small-scale adjacent plantings, and the wind effects caused by nearby buildings. These areas need to be further explored in subsequent studies.

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CRediT authorship contribution statement

Geunhan Kim: Writing – original draft. **Gunwon Lee:** Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Gunwon Lee reports financial support was provided by Korea Agency for Infrastructure Technology Advancement (KAIA). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

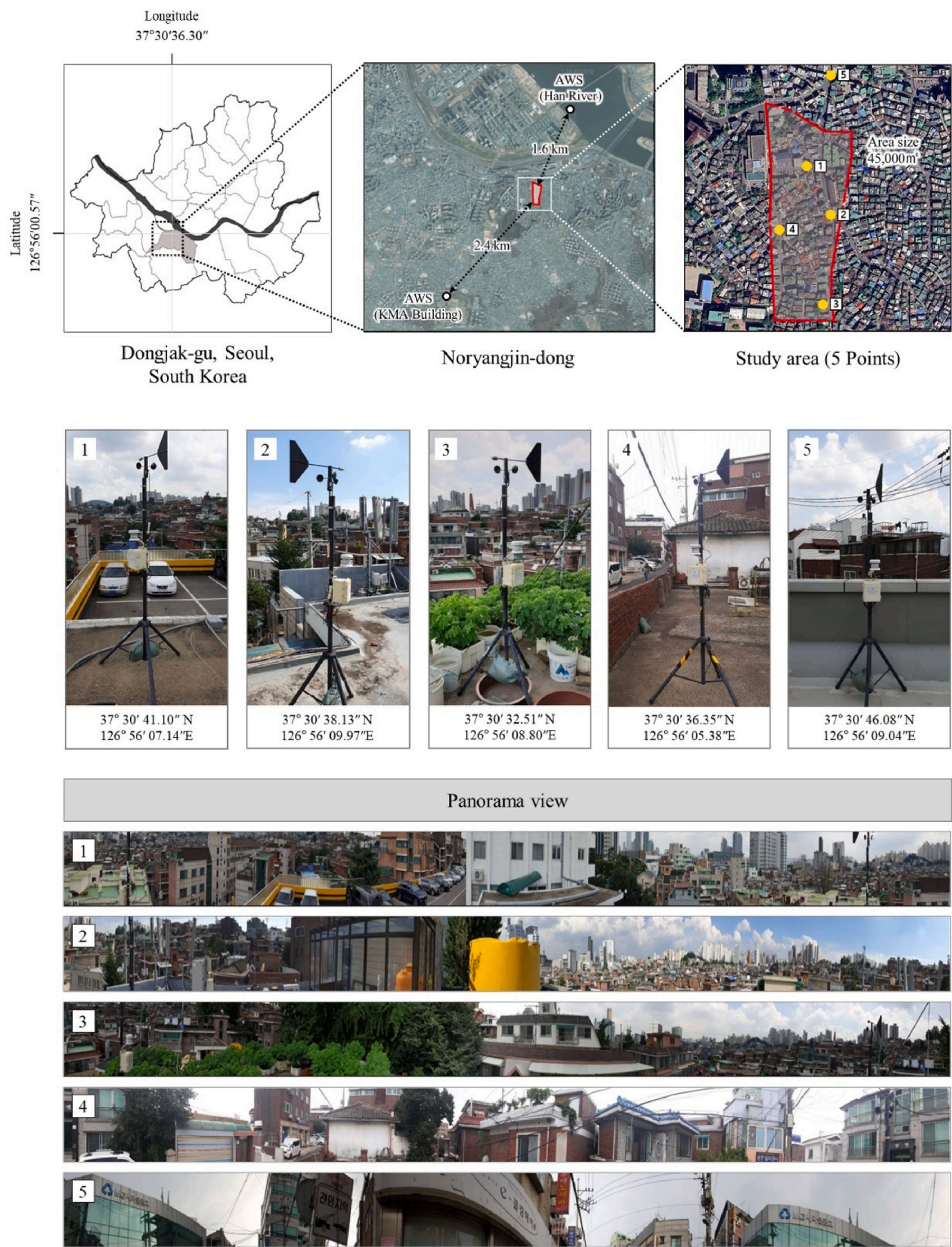


Fig. A.1. Measurement locations within the subject site and panoramic views

Table A.1
Specifications of the Measurement Sensors

Sensor		Product Number	Measurement Scope	Accuracy	Manufacturer
Temperature-humidity sensor	Thermometer	S-THB-M00x	−40–+75 °C	<±0.2 °C	ONSET
	Hygrometer	S-THB-M00x	0–100 %	±2 %	
Anemometer	Wind Speed	S-WCA-M003	0–60m/s	±0.5m/s	
	Wind Direction	S-WCA-M003	0–360°	<±3°	
Data logger		HOBO U30	–	–	

Table A.2
The Parameters Necessary to Define Urban form of the Measurement Point

Point	Latitude-longitude coordinates						Height
	Latitude	Longitude					
1	37° 30′ 41.10″ N	126° 56′ 07.14″ E					8m
2	37° 30′ 38.13″ N	126° 56′ 09.97″ E					11m
3	37° 30′ 32.51″ N	126° 56′ 08.80″ E					11m
4	37° 30′ 36.35″ N	126° 56′ 05.38″ E					2m
5	37° 30′ 46.08″ N	126° 56′ 09.04″ E					8m
	Site	Point 1	Point 2	Point 3	Point 4	Point 5	
Linear meter of streets	1,209.5	54.3	72.8	68.3	66.3	118.6	[74]
No. of Blocks	11	2	5	3	3	4	
No. of Intersections	27	1	3	0	1	1	
Road area	5,899.4	115.9	632.1	186.3	293.4	784.3	[75]
Land-use	Residential	Residential	Residential	Residential	Residential	Residential, Commercial	
Land-use area	39,101	1,847	1,330	1,776	1,669	1,178	
Building Coverage Ratio	37.30	49.10	45.20	34.70	35.20	59.50	[76]
Floor Area Ratio	105.37	157.12	140.12	52.05	77.44	202.30	
Building Height	6.66	9.90	8.90	4.30	6.50	10.10	
Green Coverage Ratio	0.033	0.019	0.010	0.013	0.015	0.008	
Distance between Building	–	4.7	4.1	3.9	4.5	5.1	
Canyon Aspect Ratio	–	1.13	1.23	0.92	0.88	1.121	
Envelope Ratio	87,624.5	3,581.0	2,332.5	1,506.0	2,130.8	2,349.7	
SVF	–	0.09	0.29	0.07	0.04	0.07	
NDVI	–	0.0326	0.0701	0.1658	0.1184	0.0503	[77]
NDBI	–	0.0061	0.1104	0.0156	−0.0609	0.0848	[78]

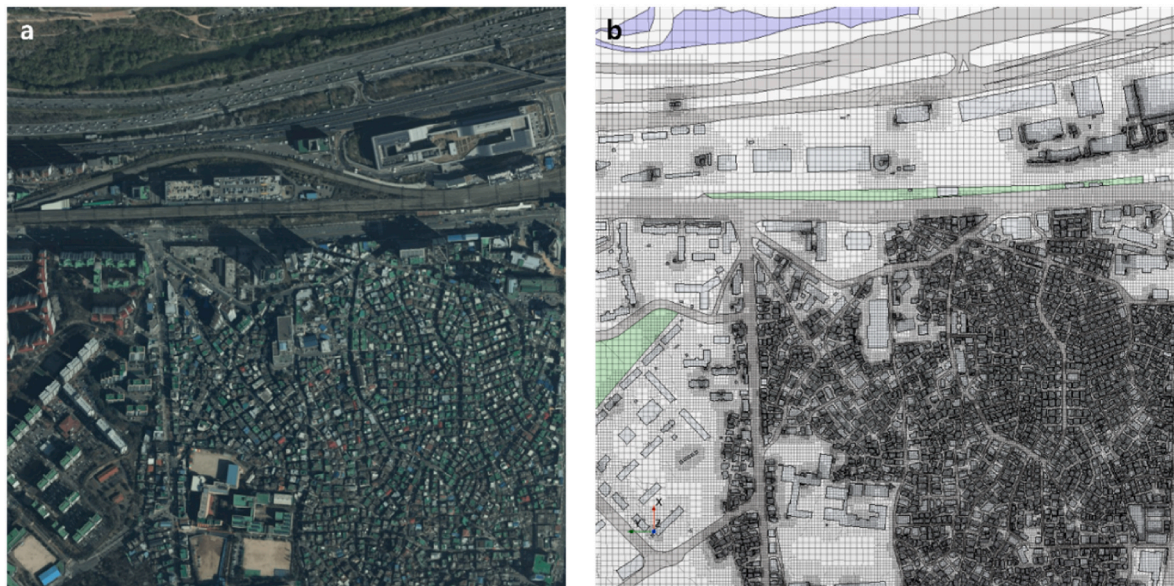


Fig. A.2. (a) Top view of the region of interest (Naver Maps); (b) Computational grid with 13,994,426 cells

Appendix B

Table B.1
Boundary Roughness and Shell Thickness Settings in the Computational Domain.

Boundary	Roughness Height [m]	Shell Thickness [m]
Green	0.25	0.03
Asphalt	0.01	0.5
Road	0.01	0.3
Buildings	0.01	0.3

Table B.2
Radiation and Thermal Properties of the Boundary Materials in the Region of Interest

Boundary	Material	Emissivity	Reflectivity	Density [kg/m ³]	Thermal Conductivity [W/m·K]	Specific Heat [J/kg·K]
Green	Grass	0.35	0.65	1,500	2.6	1,842
Asphalt	Concrete	0.5	0.5	2,400	1.7	882
Road	Asphalt	0.8	0.2	2,120	0.698	920
Buildings	Concrete	0.5	0.5	2,400	1.7	882
Land	Soil	–	–	1,500	2.6	1,842

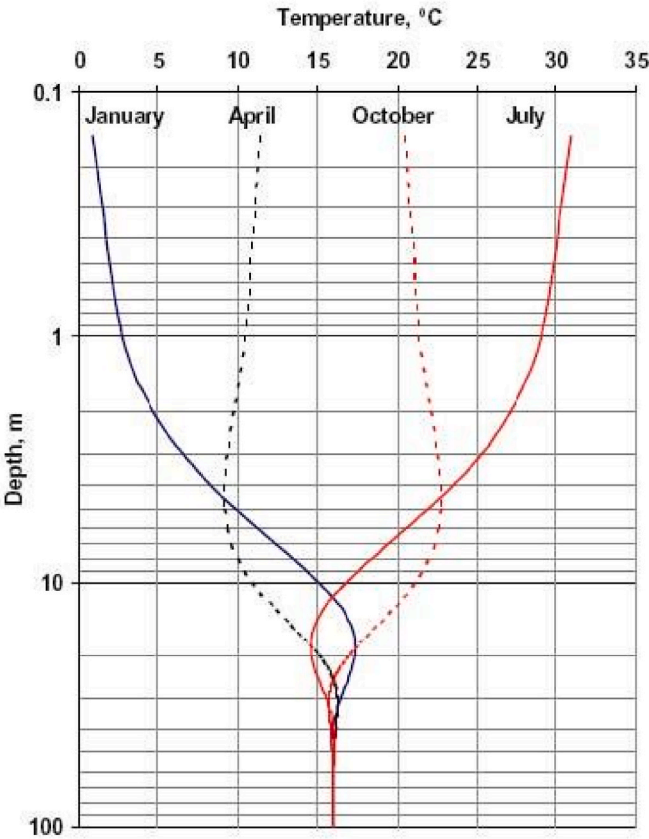


Fig. B.1. Earth temperature (°C) by depth (m)

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